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BASIC ELECTRONICS

ISCED UNIT CODE: 0714 441 04A
TVET CDACC UNIT CODE: IT/CU/ICTA/CC/01/5/MA

CONTINUOUS ASSESSMENT TEST (CAT)

INSTRUCTIONS TO THE CANDIDATE

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962

1. *Answer all questions.*
2. *Write your answers clearly and legibly.*
3. *Show all working where calculations are required.*
4. *Where diagrams are required, use appropriate electrical/electronic symbols.*
5. *The marks allocated are shown in brackets.*

1. A trainee electrician is required to assemble a simple electrical circuit for a workshop lamp. The circuit must provide a complete path for current to flow and allow the lamp to be controlled.

a) Define an electrical circuit. (2 marks)

b) Identify four components required to construct the circuit. (4 marks)

c) State the S.I. unit for each of the following quantities used when testing the circuit:
i. EMF
ii. Current
iii. Power
iv. Energy
v. Resistance (5 marks)

d) During testing, the circuit carries a current of 2 A through a resistance of 10 Ω. Calculate the voltage across the circuit. (2 marks)

e) The technician connects the circuit to a supply that provides current which changes direction periodically. State the type of current being used. (2 marks)

2. An electrical technician is testing a heater before putting it into service. The nameplate indicates a supply voltage of 240 V, while measurements taken during operation show a current of 5 A.

a) Calculate the power consumed by the heater. (3 marks)

b) During the test, the technician records 3,600 J of electrical energy. State the electrical quantity represented by this value and its S.I. unit. (2 marks)

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c) A second component has a resistance of 48 Ω. State the electrical quantity represented by the value and its S.I. unit. (2 marks)

d) Explain the difference between AC and DC with reference to the direction of current flow. (3 marks)

3. An electronics technician is repairing a control board containing four unidentified components. The components are used for controlling current, storing electrical charge, allowing current to flow in one direction and storing energy in a magnetic field.

a) Identify the four electronic components described above. (4 marks)

b) State one characteristic of each component identified in (a). (4 marks)

c) Give one practical application of each of the four components in an electronic circuit. (4 marks)

d) A technician replaces several individual components with a single packaged device containing many interconnected electronic components. Identify this type of device and state three characteristics of it. (3 marks)

4. A trainee is investigating a semiconductor device used in a power supply. The device is made from a material whose electrical conductivity is between that of a good conductor and an insulator.

a) Identify the class of material used in the device. (2 marks)

b) Name two materials commonly used to manufacture semiconductor devices. (2 marks)

c) Explain how the movement of electrons in a semiconductor differs from that in a conductor. (3 marks)

d) During manufacture, the semiconductor material is modified to produce two different regions. Explain the difference between P-type and N-type semiconductor materials. (4 marks)

e) A technician connects a P-N junction diode to a DC supply in two different ways. In the first connection, the diode permits significant current to flow. In the second connection, only a very small current flow.

Describe the condition of the P-N junction in each connection. (4 marks)

5. A computer technician is servicing a computer whose user reports that some programs take longer to load than expected. The technician examines the different memory levels and storage devices installed in the system.

a) Explain the purpose of computer memory in a computer system. (2 marks)

b) Classify computer memory into its two main categories. (2 marks)

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c) State the function of each of the following in the computer system:

i. RAM

ii. ROM

iii. Registers

iv. Cache memory (4 marks)

d) The technician needs to arrange the following according to their speed, beginning with the fastest:

- Secondary storage
- Registers
- Main memory
- Cache memory

Arrange them in the correct order. (4 marks)

e) A computer laboratory uses semiconductor chips, magnetic disks and optical discs to store data. Identify the three memory/storage technologies represented by these devices. (3 marks)

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2a) Electrical circuit is a conducting path through which electric energy passes through from the source and back to the source (2)

b) Source
Load
Switch
Conductor (4)

c) (i) Emf - volts (V)
(ii) Current - Amperes (A)
(iii) Power - Watts (W)
(iv) Energy - Joules (J)
(v) Resistance - ohms (Ω) (5)

d) I = 2A
R = 10Ω
V = ?
V = IR (2)
V = 2 × 10 = 20 A V
20V

e) Alternating Current (AC)

f) V = 240 V
I = 5A
P = IV
P = $\frac{240}{5}$
= 48 Watts.

g) Energy - SI unit - Joules (J)
Resistance - SI unit - Ohms (Ω) (2)
h) Alternating current changes direction periodically while DC direct current transmits electrical energy in a direct direction. (2)

3a) Capacitor
Inductors
Transistor
Intergrated circuit.

3b) Inductors are electronic components that stores energy in a magnetic field. (1)

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Capacitors are electronic components that allows current to flow in one direction.

Integrated circuits are components used for controlling current.

Transistors are electronic components used for storing electrical charge.

d) Integrated circuits.

They consume low power.

They are cheap.

They are fast. (5)

4a)

b) Silicon

Germanium. (2)

c) The movement of electrons in a semiconductor moves in an atom while that in a conductor moves in form of a current.

d) The P-type semiconductor is positive while the N-type is negative

The P-type uses forward bias while N-type uses reverse bias.

e) Forwatd bias

Reverse bias (2)

5a) To store files and applications in a computer system.

b) Prim Random Access Memory (RAM)

Read only Memory (ROM)

c) i) RAM - it is used to read and write.

ii) ROM - it is used to read only.

iii) Registers

iv) Cache memory is fast and it is also used to give instructions to the CPU.

d) Cache memory ~~is fast and is also used to give instructions to the CPU.~~

Main memory

Registers

Secondary storage

e) Secondary memory. (1)

Primary memory.

Cache memory.

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